

Assignment 1: Fundamentals of Image Processing and Computer Vision

Submission date: 15th September 2025

Answer the following questions based on concepts of Image Processing and Computer Vision:

1. Explain the significance of image processing in real-world applications. Give at least two examples.
2. Describe the components of an image processing system with a neat diagram. Define digitization of images. Explain the steps involved in converting a continuous image to its discrete representation.
3. Differentiate between adjacency, connectivity, and regions of pixels with examples.
4. What are the differences between convolution and correlation operations in image processing? Provide an example where convolution is preferred.
5. Discuss noise reduction techniques in image preprocessing. Illustrate with any one filtering method.
6. Describe any three feature extraction techniques (e.g., edge detection, corner detection, texture analysis) and their applications.
7. Solve this Pixel Neighbourhoods & Connectivity

Given the binary image (1=foreground, 0=background):

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

- a) Number of **4-connected** components?
- b) Number of **8-connected** components?
- c) List boundary pixels (row, col) for the 4-connected component containing pixel (1,2) (1-indexed).

8. Solve this: Histogram Equalization (8 levels)

An 8-level image (gray levels 0–7) has the following histogram (counts):

r	0	1	2	3	4	5	6	7
n(r)	300	300	200	100	50	25	15	10

Total pixels $N = \text{sum of counts}$.

a) Compute the CDF and the **equalized mapping** $s(r) = \text{round} \left(\frac{L-1}{N} \cdot \text{CDF}(r) \right)$, $L = 8$.

b) Report the final mapping table $r \rightarrow s$.